

# Zoe Ko

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## Education

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- Johns Hopkins University** 2024 – Present  
Physics and Astronomy Ph.D. Student (current)  
M.A. Physics (2026)
- University of California, Berkeley** 2019 – 2023  
B.A. Astrophysics  
B.A. Data Science  
High Distinction in General Scholarship (magna cum laude equivalent)

## Research Experience

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- Johns Hopkins University** August 2024 – Present  
Research Assistant | Advisor: Néstor Espinoza
- Implemented a Gaussian Process-aided atmospheric retrieval framework to mitigate stellar contamination in transmission spectroscopy.
  - Applied retrieval framework to constrain the atmospheric properties of a gas giant orbiting an M dwarf to better understand giant planet formation around low mass stars.
- Lawrence Livermore National Lab, NASA Pandora Graduate Intern** March 2026 – September 2026  
Graduate Intern | Advisor: Peter McGill
- Incorporated Gaussian Processes into a stellar retrieval pipeline designed for Pandora data.
  - Validated pipeline on synthetic observations, demonstrating improved stellar parameter recovery and contamination corrections across diverse star-planet scenarios.
- California Institute of Technology LIGO Lab** June 2022 – January 2024  
Research Intern | Advisor: Katerina Chatziioannou
- Developed and implemented a hierarchical Bayesian inference framework to analyze binary black hole spin distributions using LIGO gravitational wave data.
  - Generated mock binary black hole populations with different component spin parameters to quantify how much spin information is extractable from observed effective spins.
- University of California, Berkeley SETI Research Center** June 2021 – January 2023  
Research Intern | Advisor: Howard Isaacson
- Developed a laser detection algorithm to search for potential technosignatures in optical stellar spectra and applied algorithm to over 1000 spectra of nearby stars to set detection limits.
- Pennsylvania State University** September 2019 – February 2023  
Research Assistant | Advisor: Ian Czekala
- Designed a pipeline from scratch to detect radial velocity variations through cross-correlation techniques and applied pipeline to 28 candidate binary star systems.

## Awards and Fellowships

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- William H. Miller III Graduate Fellowship** August 2024 – August 2025
- Awarded a 12-month fellowship from Johns Hopkins to support independent graduate research.
- Fulbright Scholarship Award** September 2023 – July 2024
- Designed and taught an English language curriculum for seventh and tenth grade students in Thailand.

## Publications

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- **Z. Ko**, P. McGill, B. V. Rackham, D. Apai, A. R. Iyer, J. F. Rowe, L. Welbanks, S. Saha, Y. Rotman, & The Pandora Team. 2026, "NASA's Pandora SmallSat Mission: Gaussian Processes for Simulated Stellar Contamination Corrections amid Imperfect Stellar Models," AJ, in prep.

- **Z. Ko**, N. Espinoza, A. Jordán, D. Bayliss, E. M. Bryant, M. Schlecker, M. J. Hobson, R. Brahm, R. Burn, & T. Henning. 2026, "JWST Transmission Spectroscopy of TOI-3235 b: Challenges in Constraining Giant Planet Atmospheres around M Dwarfs amid Stellar Contamination," *AJ*, 172, 174
- S. J. Miller, **Z. Ko**, T. Callister, & K. Chatziioannou. 2024, "Gravitational waves carry information beyond effective spin parameters but it is hard to extract," *Phys. Rev. D*, 109, 104036
- A. Zuckerman, **Z. Ko**, H. Isaacson, S. Croft, D. Price, M. Lebofsky, & A. Siemion. 2023, "The Breakthrough Listen Search for Intelligent Life: A Laser Search Pipeline for the Automated Planet Finder," *AJ*, 165, 114

## Conference Talks and Posters

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- **Z. Ko**, P. McGill. Characterizing Exoplanet Atmospheres with the Pandora SmallSat Mission, Space Science Institute Seminar, Lawrence Livermore National Laboratory, Livermore, CA, September 2026. (Oral Presentation)
- **Z. Ko**, P. McGill. Empirically Correcting Stellar Contamination with NASA's Pandora Mission, Annual Poster Showcase, Lawrence Livermore National Laboratory, Livermore, CA, July 2026. (Poster)
- **Z. Ko**, N. Espinoza, A. Jordan, P. McGill. JWST Transmission Spectrum of a GEMS Target, Exoplanets 6, Porto, Portugal, June 2026. (Poster)
- **Z. Ko**, P. McGill. Mitigating Stellar Contamination with Pandora Data, Pandora Team Meeting, NASA Goddard, Greenbelt, MD, June 2026. (Oral Presentation)
- **Z. Ko**, N. Espinoza, A. Jordan. Probing the Atmospheres of Giant Planets around M Stars Amid Stellar Contamination, Invited Speaker, Universidad de Chile, Santiago, Chile, March 2026. (Oral Presentation)
- **Z. Ko**, N. Espinoza, A. Jordan. Characterizing Giant Planets around M Dwarfs amid Stellar Contamination with Gaussian Processes, Invited Speaker, Pontifical Catholic University, Santiago, Chile, March 2026. (Oral Presentation)
- **Z. Ko**, N. Espinoza, A. Jordan. Unveiling the Atmospheres of Giant Planets around M Stars Amid Stellar Contamination, Invited Speaker, Universidad Diego Portales, Santiago, Chile, March 2026. (Oral Presentation)
- **Z. Ko**, N. Espinoza, A. Jordan. Mitigating Stellar Contamination in JWST Transit Data with Gaussian Processes, Exojamboree, Baltimore, MD, November 2025. (Oral Presentation)
- **Z. Ko**, N. Espinoza, A. Jordan. A Gaussian Process Framework for Exoplanet Atmospheric Retrievals, Exoclimates VII, Montreal, QC, Canada, July 2025. (Poster)
- **Z. Ko**, N. Espinoza, A. Jordan. Gaussian Process Retrievals for TOI-3235b, Emerging Researchers in Exoplanet Science Symposium X, Princeton University, Princeton, NJ, June 2025. (Oral Presentation)
- **Z. Ko**, S. J. Miller, K. Chatziioannou. Analyzing the Effective and Component Spin Distributions of Binary Black Hole Mergers, American Physical Society April Meeting, Minneapolis, MN, April 2023. (Oral Presentation)
- **Z. Ko**, H. Isaacson, A. Zuckerman, S. Croft. Search for Laser Emission Lines with the Automated Planet Finder Telescope, American Astronomical Society Meeting, Seattle, WA, January 2023. (Poster)
- **Z. Ko**, H. Isaacson, A. Zuckerman. Search for Laser Emission Lines with the Automated Planet Finder Telescope, AbSciCon, Atlanta, GA, May 2022. (Poster)
- **Z. Ko**, H. Isaacson, A. Zuckerman, S. Croft. Search for Laser Emission Lines with the APF, Order of the Octopus, Virtual, August 2021. (Poster)

## Skills

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**Programming:** Advanced proficiency in Python (NumPy, SciPy, pandas, matplotlib, Astropy, etc.), Java, R.

**Statistical Modeling:** Extensive experience with Bayesian inference frameworks (PyMC, emcee, dynesty) and Gaussian process modeling (tinygp, george, celerite).

**Languages:** Native fluency in English and Mandarin; strong proficiency in Spanish and Thai.

## Teaching and Outreach

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**STEM Mentor at learnSTEM Research Mentorship Program** March 2026 – June 2026

- Mentor a high school student in a 9-week program to complete an independent research project.

**Physics and Astronomy Peer Mentor at Johns Hopkins** August 2025 – May 2026

- Advise physics and astronomy undergraduate students academically and professionally.

**Peer Tutor at the Berkeley Student Learning Center**

August 2022 – May 2023

- Provided weekly tutoring services for introductory astrophysics courses.

**Research Mentor at the Berkeley Undergraduate Lab**

August 2021 – May 2022

- Mentored five undergraduates through a year-long independent research project on binary star systems.

**Student Mentor and Outreach Director of Berkeley Engineers and Mentors** January 2020 – May 2023

- Led weekly science labs at elementary and middle schools in Oakland and coordinated outreach events.